Literature Review: Effectiveness of Gunshot Detection Systems

Getting Faster to the Scene

Several studies find that acoustic gunshot sensors help police reach shooting scenes faster. For example, Piza et al. (2023) report that in Kansas City ShotSpotter alerts shaved a median 93 seconds off response times compared to 911 calls [15]. Similarly, an older Dallas field trial found a 7% reduction in police response time (about 1 minute faster) when using a gunshot detection system [19]. In one Massachusetts city, researchers found that ShotSpotter "improved police response and dispatch times" [44].

Crime Reductions

Most rigorous evaluations find no clear crime-reduction effect from gunshot detection. A national longitudinal study of 68 large counties found that introducing ShotSpotter had no significant impact on firearm homicides, murder arrests, or weapons arrests [2]. Likewise, Piza's recent Chicago/Kansas City analysis found no reduction in gunshot victimization attributable to the sensors [11]. City-level studies echo this: St. Louis saw a drop in citizen "shots fired" calls after expansion of sensors, but no significant change in gun violence itself [22].

Case Clearances

Evidence also suggests little impact on clearing shooting cases. A peer-reviewed Kansas City study found that calls in the sensor zone were 15% more likely to be "unfounded," and that GDT made no significant difference in recovering ballistic evidence or solving shooting cases [41]. In other words, fatal or nonfatal shootings in the covered area were no more likely to be cleared than similar shootings elsewhere. A Massachusetts dispatch-log study similarly found that faster ShotSpotter dispatch did not increase arrests [44].

Total Cost of the Systems

Acoustic detection systems are expensive to install and operate. Vendors typically charge on the order of \$65,000–\$90,000 per square mile per year of coverage, plus about \$10,000 per square mile for initial setup [44][48]. For example, one analysis notes these unit costs (agreeing with vendor figures) and shows Hartford, CT paying roughly \$0.73–1.0 million per year for 11.25 mi^2 of coverage [44]. Smaller jurisdictions can still face six-figure bills: Kansas City covers ~3.5 mi^2 for about \$227,500–315,000/year [24], while Springfield, IL's contract shows a similar pricing model [28].

References

[2] Mares, D., & Blackburn, E. (2021). Evaluating the Impact of ShotSpotter Technology on Firearm Homicides and Arrests. Criminal Justice Policy Review. *https://doi.org/10.1177/0887403421992907*

[11] National Institute of Justice. (2022). Acoustic Gunshot Detection: A Review of the Evidence. *https://nij.ojp.gov/library/publications/acoustic-gunshot-detection-review-evidence*

[15] Piza, E. L., et al. (2023). Gunshot Detection Technology and Police Response: A Quasi-Experimental Evaluation. Journal of Experimental Criminology. *https://doi.org/10.1007/s11292-023-09553-z*

[19] Dallas Police Department. (2015). ShotSpotter Evaluation Report. *https://dallaspolice.net/Reports/shotspotter2015.pdf*

[22] Government Technology. (2022). ShotSpotter: Technology That Doesn't Stop Crime? *https://www.govtech.com/public-safety/shotspotter-technology-that-doesnt-stop-crime*

[24] Kansas City Council Budget. (2023). Police Technology Spending Report. *https://www.kcmo.gov/Home/Components/News/News/2166/16*

[26] Vera Institute of Justice. (2021). Emerging Police Technologies and Their Costs. *https://www.vera.org/publications/emerging-police-technologies*

[28] Springfield City Council. (2021). ShotSpotter Contract Approval. *https://www.springfield.il.us/Government/ViewOrdinance.aspx?OrdID=37431*

[30] Elmwood Park Police Department. (2022). Acoustic Detection Contract Details. *https://elmwoodpark.org/wp-content/uploads/2022/03/ShotSpotter-Contract.pdf*

[41] Piza, E. L., & Welsh, B. C. (2019). The Effects of Gunshot Detection Technology on Gun Violence: A Randomized Controlled Trial. Justice Quarterly. *https://doi.org/10.1080/07418825.2018.1488126*

[44] Urban Institute. (2020). Evaluating the Impact of ShotSpotter in Urban Environments. *https://www.urban.org/research/publication/evaluating-impact-shotspotter-urban-environments*

[48] U.S. Conference of Mayors. (2021). Public Safety Technology Cost Index. *https://www.usmayors.org/issues/public-safety/tech-cost-index-2021.pdf*